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"Except where explicitly stated, all work in this report, is my own original work and has not been submitted elsewhere in fulfilment of the requirement of this or any other award"

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Date:
March 30, 2026

Abstract

Throughout the rise of social media and smartphones, an ever-growing topic of discussion has been the issue of individuals developing an overdependence on said technologies, referred to as "Social Media Addiction" and "Smartphone Addiction" respectively. Research into the topic discussing the potential causes for it, and possible methods to treat the issue have been performed extensively in the past, with the latter looking towards psychological-based treatments such as Cognitive-Based Therapy and Reality Therapy, and in recent years, some looking at mobile application that use some form of usage tracking to address the issue. Research has shown that psychological-based treatments has provided promising results in treating similar issues such as anxiety, and has even seen integration with mobile apps, with it also showing viability when addressing other mental health concerns. Similarly, studies conducted show viability in utilising usage tracking when addressing the issue, particularly when integrated into a mobile app. However, little work has been done to utilise both of these concepts together into a single mobile app.

This project showcases an Android-based mobile application that integrates concepts from Reality Therapy alongside usage tracking technologies, with the overall objective being to lower an individuals usage in an attempt to address the issue of Social Media and Smartphone Addiction. To evaluate the project, a small group of participants were given the application to use over a three-week period, and were asked to send their usage data each week during this time. By the end of this period, they were also provided a survey where they could provide their own opinions on the app.

Results from the evaluation show that

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Abbreviation / Acronym	**Description / Definition**
RT	Reality Therapy
WDEP	Wants, Direction, Evaluation, and Planning & Commitment
SRS	Software Requirements Specification
PRI	Primary Research Instrument
SDLC	Software Development Lifecycle
SA	Smartphone Addiction
SMA	Social Media Addiction
LSEP	Legal, Social, Ethical, and Professional
QS	Question Set

Table 1: List of Abbreviations and Acronyms

1 Introduction

This section will provide a brief overview on smartphones and social media. It will then provide context on the issue of smartphone addiction and social media addiction, including how people develop such addictions, and the effects they have on an individual. Existing apps and studies that attempt to deal with the issue are also mentioned. It will also provide an overview on the project and it's objectives to be achieved in the literature & technology review and during development of the app.

1.1 Project Background

Social media (SM) and smartphones have been adopted as an important piece of daily life for a substantial number of people in the world. They have provided us the ability to connect with from around the globe on a never-before-seen scale, allowing for friendships, communities, and relationships to be built from virtually anywhere. Dating back MySpace in the mid-2000s, the release of Apple's "iPhone" and the release of the HTC Dream running the "Android" operating system in 2008, to SM such as Twitter and Facebook reaching out to billions around the globe, smartphones and social media looks to be omnipresent within a majority of peoples' lives, with it being estimated that 1 in 3 people globally have some form of SM presence (Shah 2021), reportedly 85% of adults in the United States own a smartphone (Ouellette and Rao 2022).

Over the years, worries have been raised regarding many issues that concern smartphones and SM, particularly surrounding the problem of individuals developing an overdependence on their smartphones and SM that negatively affects their day-to-day life (Allen and Buzzi 2020; Primack et al. 2017). Commonly categorised as "Smartphone Addiction" (SA) and "Social Media Addiction" (SMA) and considered subsets of behavioural addiction (King and Dong 2017; Hou et al. 2019), the issue has been largely discussed and researched over the last 15 years, with researchers looking into potential causes and correlations with other health effects.

Bhalerao et al. (2020) analyzed SA and provided potential causes behind SA, one of which being how the current generation has been brought up alongside the internet and technology, therefore possibly feeling a preference to communicate over the internet instead of in-person, fueling the dwindling need for face-to-face communication. King and Dong (2017) made a series of hypotheses in relation to college students and the effects related to SA, looking at SA's associations with "life stressors" (a measure of stress and anxiety), "digital overload" (a measure of information and communication overload), retention, academic performance, and social well-being.

The results found that SA could be positively correlated with life stressors, digital overload, retention, and academic performance, something consistent with previous studies.

Potential treatments have been researched as well. A China-based study from 2019 analyzing SMA (Hou et al. 2019) developed a cognitive-behavioural-therapy-based intervention program that used various techniques including reminder cards, journal entries before sleeping, and cognitive reconstruction. The results showed that compared to the control group, those who took part in the program saw improvements in mental health, engagement in learning, sleep quality, and also saw a drop in SM usage.

The "AntiSocial" app, developed by Australian company BugBean for Android devices allowed users to restrict access to certain apps and provided app usage statistics, but also added other functionality like usage comparisons with others that were within their age group, location, and occupation. It also showed users how many times they picked up their device, and how long they used certain apps for (Michael 2017). The author stated that the app ranked 3rd in downloads on the UK Google Play store mere days after launch, "demonstrat[ing] the need for [AntiSocial.io]" (ibid.).

However, there seemingly exists a gap of intervention apps that deal specifically with SA or SMA, and that also allow self-reflection from the user, and most literature on the topic mostly provide analysis on apps that deal with adjacent issues. A 2021 study by Oliveira et al. (2021) reviewed the effectiveness of psychological intervention mobile apps, and while they concluded that the apps showed promise, none of the literature specifically looked at addressing the issue of SA or SMA.

1.2 Project Overview

1.2.1 Project Outline

Evidence has shown that smartphone apps that help with SA and SMA are in-demand, and significant amounts research have been conducted suggesting treatments to the issue. However, most consumer products on the app store opt to utilize app-restriction features. Because of this approach, young adults who use these apps aren't given much opportunity to reflect on the nature of SA & SMA and the potential causes of it, leading to very little character growth for the end-user. However, by taking a different approach by integrating psychology, young adults can be given the opportunity to reflect on their addiction and learn how to address the

issue. With that being said, the research statement is as follows:

To create a mobile application that integrates psychological intervention methods alongside usage tracking technologies to help address smartphone and social media addiction

1.2.2 Project Objectives

The primary objective of this project is to create an application that can guide the user into reducing their smartphone & social media usage with the use of usage tracking and psychological methods.

Secondary Objectives

Investigate existing mobile apps that use psychological methods to address mental health issues
Research into how existing mobile applications make use of psychological methods when addressing issues such as depression and anxiety to understand how viable using mobile apps are to address similar issues.

Investigate the contextualization of SA and SMA in existing apps
Research how the issue of SA and SMA are contextualized within existing intervention apps and what issues could arise from how they are presented. This will help in enabling growth in a user when it comes to learning about SA and SMA.

Investigate psychological frameworks
Research how psychological methods have been used in the past, specifically to treat SA or SMA, and see if such methods have been integrated with mobile apps to gain a further understanding on how these methods could be integrated or improved upon.

Investigate methods to track app usage
Research into how apps currently track phone activity, and how they can be utilized in the context of the project, as a major component of the application makes use of usage tracking.

Primary Objectives

Develop the smartphone intervention app
Create the application based on the knowledge gained from the literature and technology review. This will involve integrating the chosen psychological framework and usage tracking method before testing the app's functionality via test cases to ensure

the app functions as intended.

Gather participants for evaluating the app and perform evaluation
Participants will be used when evaluating the developed application. They will be given an overview of the app before being given a consent form to sign to consent to participating in the research.

Analyze the results and obtain a conclusion
Collect results from the evaluation and see if the app was impactful at reducing smartphone and social media usage. Then evaluate based on the results, and see if the problem statement and hypothesis were met.

1.3 Project Hypothesis

Testing the hypothesis will involve providing participants with the developed application for a period of three weeks. During this period, participants will attempt to reduce their smartphone & social media usage by using the developed mobile application with usage tracking and psychological intervention methods. Given this, the hypothesis for this project is:

The combination of usage tracking and intervention methods rooted in psychology within the created mobile application will help to address and reduce smartphone & social media usage in young adults.

1.4 Aims and Methodology

1.5 Structure of the Report

Before continuing with the remainder of this paper, a brief overview of the structure of the report is provided.

Chapter 2 - Literature and Technology Review - This chapter provides an extended review of the current literature surrounding topic areas relevant to the report while also analysing technical methods utilised by apps with a similar intention to the PRI outlined within the primary objectives.

Chapter 3 - Execution - This chapter provides an extensive look towards the development of the PRI, providing commentary towards the research methodology used, and discussion and justification on the design choices, technological components used, and features implemented when developing the PRI.

Chapter 4 - Evaluation and Discussion - This chapter presents a in-depth discussion of the results of the evaluation, including the findings themselves, reasoning on the findings, and initial conclusions from the findings.

Chapter 5 - Legal, Social, Ethical, and Professional Issues - This chapter gives commentary on any Legal, Social, Ethical, and Professional the project encountered over its lifespan, with additional discussion provided towards the Sustainable Development Goals the project aligns with when discussing social issues.

Chapter 6 - Conclusions, Project Critique, and Further Work - This chapter concludes the project by summarizing the project in it's entirety up to this point, before then presenting a final answer to the project's hypothesis. It then continues by providing critical analysis of the project and it's limitations, and finally, discusses what further work the project may lead in the future.

2 Literature and Technology Review

This literature and technology review will review mobile apps that use psychological methods to address mental health issues before looking at how existing apps provide context regarding SA & SMA and the issues associated with it. Potential psychological frameworks for treating SA & SMA are reviewed before researching how app usage can be tracked.

2.1 Investigate existing mobile apps that use psychological methods to address mental health issues

The project background mentioned that although researchers have studied SA and SMA, few apps are available, or have been created for research purposes that handle the issue. However, apps that address similar issues exist and have extensively been reviewed and evaluated. This section reviews existing studies on these apps to gain a further understanding on the viability of treating mental health issues.

2.1.1 Prior research on mobile app interventions

Researchers Weisel et al. (2019) reviewed mobile apps that integrated psychological frameworks to treat mental health issues such as depression, anxiety, drug use, among others, with their intention being to evaluate the effectiveness of these apps in reducing mental health symptoms. The review discusses features used, with the most common being behaviour monitoring, participant engagement via explicit input, tailoring based on feedback, and gamification, while also noting the majority of apps opted to use CBT as their psychological framework. They discover that out of all the mental health issues targeted, depression, sleep-related issues, and smoking returned a significant effect, with the review stating that "our knowledge on how to design effective mental health apps is very much at the beginning".

They continue by suggesting different approaches that focus more on the user and the framework, as well as engagement with the app via usage patterns, before saying that at the current point in time, there isn't much significant evidence that supports using mobile apps for treating mental health as an alternative to traditional treatment, highlighting that while apps targeting depression, sleep issues and smoking showed promising results, the results showed a high risk of bias within them (ibid.).

Oliveira et al. (2021) reviewed mobile app-based interventions utilizing psychological frameworks targeting college students, aiming to review the outcomes of using

app-based psychological interventions instead of traditional therapy, as well as looking how these apps are being embraced by students. The review details the types of mental health issues these apps target, while also noting that a majority of these apps are "self-guided" in nature (Oliveira et al. 2021). Psychological frameworks used are mentioned, with Cognitive Behavioural Therapy (CBT) being the main framework used or combined with another framework, with other frameworks such as Acceptance and Commitment Therapy and Self-determination Theory mentioned.

The review then concludes by saying that the current literature demonstrates that using mobile apps for mental health intervention is a viable option and could be a great resource for counselling services, noting that future work into the topic should utilize alternative psychological frameworks, look further into what students and therapists would want out of similar apps, and that future studies should stray from monetary incentives for students looking to participate (ibid.).

2.1.2 Discussion

Weisel et al. (2019) found that while apps that targeted depression, sleep-related issues, and smoking showed some results, the field was merely in its infancy when it came to designing effective applications that integrate psychological frameworks. Later, Oliveira et al. (2021) showed that mobile apps that targets mental health issues aimed towards college-level students were a viable alternative option for treating mental health issues due to student embracement, showcasing growth within the topic between the two points in time, though noted that this could've been due to incentives attached within the latter study.

Despite lacking insight on psychological frameworks other than CBT, showing a need to expand the types of psychological frameworks used for these apps, and the use of incentives which may bias user retention, the studies demonstrate that the viability of a mobile mental health app using psychological frameworks has grown, and that developing such app is doable, as long as aspects such as other psychological frameworks, and user incentive are considered. The studies also highlight that while apps such as the ones discussed are in their infancy and may not deliver the same results as traditional treatment, the ease of access compared to traditional treatment provides another point to its potential viability.

2.2 Investigate the contextualization of SA and SMA in existing apps

The project background mentioned that intervention apps are developed with mostly functionality in mind, while neglecting any sort of teaching element, therefore leaving

the user little to no room to learn about SA and SMA. This section reviews how SA and SMA are contextualized, and how this could be improved.

2.2.1 Issues with contextualization

Dosselaar (2017) puts forward an argument stating that SA apps created by developers aren't suitable for treating SA due to developers having limited knowledge on the topic of SA, and points out the irony in attempting to treat SA with the use of a smartphone. This sentiment is shared in Petrucco (2021), stating that proposing technology as a solution to an issue caused by technology itself is paradoxical in nature.

Dosselaar says that developers lack a deep understanding of the topic, and points out the overall complexity of the issue, mentioning how SA is a product of many other factors; not just the phone, and the apps simplify what is often a complex issue by acting as a one-size-fits-all solution (Dosselaar 2017; Roffarello and Russis 2019). They also fail to consider context behind smartphone usage, noting how it can be productive (Dosselaar 2017) and non-disruptive depending on where it's used (Petrucco 2021). Another issue is that the apps lack any kind of opportunity for the user to understand and reflect on the issue of SA, noting that these apps frame SA as an issue that can be quantified, while lacking any features that offer information about SA (Dosselaar 2017), something that plays a role in treating the issue (Rosa, Stracuzzi, and Faro 2022), and could influence app effectiveness (Rahmillah et al. 2023).

The author concludes by stating that SA apps are limited in effectiveness as they focus on practical functionality of the app rather than the contextualization and opportunity for reflection for the user, before recommending researching integrating psychological frameworks into the apps (Dosselaar 2017).

2.2.2 Discussion

While it can be argued that some of these issues have been addressed, for consumers, most of these issues are yet to be addressed by anyone but the developers of these apps (ibid.). These apps focus only on features such as self-monitoring and app restriction (Löchtefeld, Böhmer, and Ganev 2013), rather than providing opportunity for end-users to reflect and learn. These apps frame SA and SMA as an issue that can be solved with the apps themselves, rather than it being a complex issue (Santhosh 2025), emphasised by the fact that not all smartphone usage is the same for each person (Petrucco 2021), bringing suggestions of incorporating aspects such

as psychological frameworks into these apps to make them more effective (Dosselaar 2017).

2.3 Investigate psychological frameworks

The project background mentioned that plenty of studies have been conducted utilising different psychological methods to treat SA and SMA. This section aims to review the most common psychological methods to understand which method would work best with the project.

2.3.1 Existing psychological frameworks for SA and SMA

The previous section highlighted the need for research into how psychological frameworks could be integrated into consumer SA and SMA intervention apps to make them more effective, a sentiment shared with Oliveira et al. (2021), who mentioned future intervention apps should incorporate [other] psychological methods.

Intervention apps developed for research purposes commonly use Cognitive Behavioural Therapy (CBT) as their psychological framework (Weisel et al. 2019; Oliveira et al. 2021), though other frameworks such as mindfulness, reality therapy, and theory of planned behaviour have also been utilized to mixed success (Gajecki et al. 2014; Witkiewitz et al. 2014; Lu et al. 2025).

2.3.2 Cognitive Behavioural Therapy

Cognitive Behavioural Therapy (CBT) is a psychological framework that take an "active, problem-focused, time-sensitive treatment approach" aiming to increase adaptive behaviour and reduce emotional distress in an individual with mental health issues via targeted, short-term strategies that intends to alter maladaptive emotional responses in an individual by changing their thoughts and/or behaviours (Wenzel 2017; Kaczkurkin and Foa 2015). Despite results lacking in some areas, it can be argued that it's the framework with the highest usage (Hofmann et al. 2012) overall, with some considering it the "gold standard" (Beck 2020) due to the proven success the framework has had on treating certain issues such as nicotine overdependence, schizophrenia, depression, and anxiety (Hofmann et al. 2012).

The main CBT techniques used in intervention apps are behavioural activation (A focus on engaging a patient in positive activities and helping with aversive control (Tindall et al. 2017)), cognitive restructuring (A process that assists patients in identifying and evaluating negative thoughts (Wenzel 2017)), and problem solving (Discover solutions to a patient's issue (Denecke, Schmid, and Nüssli 2022)).

Throuvala et al. (2020) assessed if these techniques could reduce smartphone usage via three separate apps which provided these techniques (AntiSocial, Headspace, Pacifica) and found that those in the experimental group saw their smartphone usage drop and were made more aware of their usage when compared to the control group.

2.3.3 Reality Therapy

Reality Therapy (RT) is a psychological framework that can be traced back to choice theory, which theorizes that all human behaviour is a result of choices, (Bradley 2014; Wubbolding 2016). RT's primary aim is on focusing an individual's behaviour in the present by using the Wants, Direction, Evaluation, and Planning & Commitment System (WDEP system) (Bradley 2014). This involves asking an individual to define a goal for themselves, what they're currently doing, and if that behaviour is halting progress towards their goal or not, before being encouraged to replace it with a healthier alternative (Yao et al. 2017). RT has shown effectiveness in treating issues such as Internet Addiction and Internet Gaming Disorder (Yao et al. 2017; Zajac et al. 2017), though has had limited research (Bradley 2014) when treating SA and SMA compared to CBT.

Rad and Ahmadi (2018) developed a mobile app integrating RT. The app was hypothesized to reduce anxiety, SM usage, among others, and was developed for Android devices. It implemented teachings about SMA, as well as app restriction functions, with the results being that anxiety and SM usage fell when compared to the control group.

2.4 Investigate methods to track app usage

The project background mentioned that existing SA and SMA intervention applications mainly make use of app-restriction and usage tracking technologies, and due to usage tracking being a large component of the project, it would be useful to know the technological methods used for app usage tracking.

2.4.1 Justification for app usage tracking

App usage tracking has proven to be successful in reducing smartphone usage. a study by researchers Rahmillah et al. (2023) reviewed the effectiveness of apps intended to reduce smartphone use, and found that in the 13 apps reviewed, 4 apps successfully reduced smartphone usage, those of which included app usage tracking.

2.4.2 Methods of app usage tracking

Rosa, Stracuzzi, and Faro (2022) hypothesized that individuals with high levels of smartphone usage show more difficulty in not using their smartphone when compared to those with lower levels, verifying their hypothesis by developing an application called "SocialStatsApp" to track app usage of participants.

The app was developed for devices using Google's Android operating system. The researchers mention how making the app compatible for Apple's iOS devices wasn't possible as the OS disallows the collection of usage statistics for privacy reasons (Apple-Support n.d.). The programming language used was Java, one of three officially supported languages via the Android SDK, the other two being Kotlin and C++ (Android-Developers n.d.), though the study doesn't detail the reasoning behind using Java. The app makes use of the Android class android.app.usage.UsageStatsManager from the Android API which gives access to phone usage stats and history. Because of this, the app needs permission manually granted by the user. The results of the study showed that those with high levels of smartphone usage did experience more difficulty in not using their phone.

Bye et al. (2024) analyzed the effects technology has on the mental health of young people, developing an app called the "3S-YP" app that constantly collected "smartphone metadata" from participants. While they don't detail the development of the app, they mention that smartphone metadata was collected using the android.app.usage package from the Android API, while also mentioning that metadata collection was not performed for the iOS version of the application due to "data protection restrictions", a sentiment shared by the previous study, showing that developing for Android allowed for additional functionality not possible with iOS.

Another method of tracking a user's app usage is by making use of screenshots and digital well-being features included with smartphones. Ubochi (2019) investigates the use of digital health apps and their effectiveness via Apple's Screen Time feature. Due to the aforementioned privacy restrictions, the researcher instructs participants to instead screenshot multiple aspects related to the Screen Time feature. A limitation of this method however is that screenshots must be taken at a specified time, as mistiming it could yield less accurate metrics (*Mobile sensing in psychology: Methods and applications* 2024).

2.4.3 Discussion

Both studies chose Android as their development platform of choice, mentioning that iOS apps don't allow for app usage tracking due to privacy restrictions (Rosa, Stracuzzi, and Faro 2022; Bye et al. 2024; Apple-Support n.d.). This is elabo-

rated further within *Mobile sensing in psychology: Methods and applications* (2024), mentioning including iOS users would require a "jailbroken" phone, which brings a variety of issues. Meanwhile, Android allows for this as long as the user permits it. Only one of the studies makes mention of a programming language (Rosa, Stracuzzi, and Faro 2022), that being Java, and while both studies make use of the android.app.usage package within the Android API, only one (ibid.) explicitly uses the UsageStatsManager class. Given this, it is clear that the OS that the app should be developed for is Android, and that using the android.app.usage package would be a viable method of app usage tracking.

3 Execution

This section will provide an in-depth look at the process behind building the PRI. This section will include a brief analysis on the research methodology used, discussion on the described problem's characteristics, and then a extensive look at the creation of the PRI, including commentary and justification behind design choices, technologies used, and the features implemented. When necessary, this section will also reference the literature review.

3.1 Problem Discussion and Analysis (title pending)

The aim of this project is

3.2 Research Methodology

The research methodology chosen for this project was a "develop and test" methodology. This methodology was appropriate for the objective as the primary objective of this project as stated in section 1.2.2 is to:

...create an application that can guide the user into reducing their smartphone & social media usage with the use of usage tracking and psychological methods.

and ensuring the objective is met involves testing the developed application with a small group of users, and evaluating based on captured data if the PRI meets the project's main objective, research statement, and hypothesis.

3.3 Lifecycle

Early in the planning stage of the project, the project intended to make use of an iterative development methodology, more specifically, the agile methodology, due to its focus on flexibility, adaptability to any changes in requirements, and frequent use in mobile app development (Chandarana, Gada, and Kukkadapu 2024). However, during the requirements analysis stage of the project, a change to a Hybrid methodology was made. This model aims to let development teams combine approaches from multiple methodologies and tailor the Software Development Lifecycle to fit their needs (Hossain 2023). In this case, approaches from iterative and waterfall methodologies were combined to create a methodology that fit the needs of the project at that point in time. specifically, the waterfall methodology was used for the Requirements Analysis, Design, and Evaluation stage, while the iterative methodology was used for the Implementation and Testing stage. This allowed for requirements to be "fully" defined and for designs to be as detailed as possible, while also allowing for

the creation of multiple prototypes and for issues to be identified and solved during the creation of each prototype.

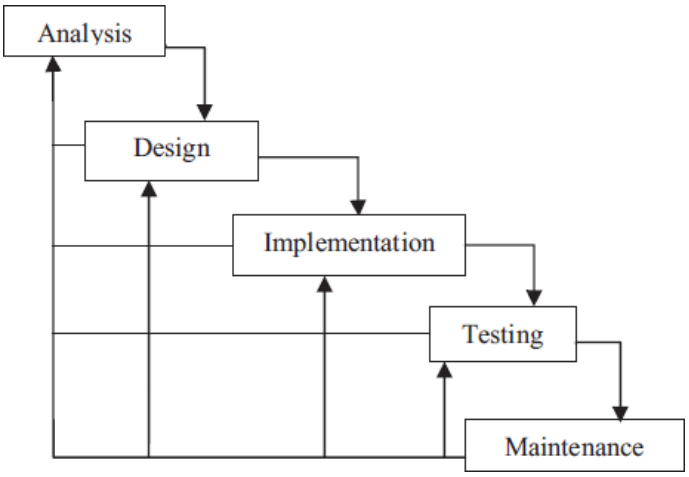
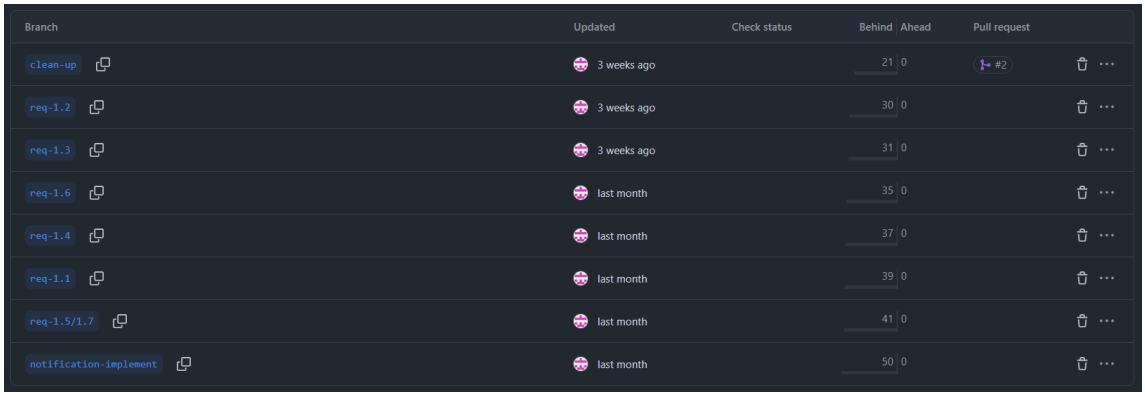


Figure 1: *Typical iterative Waterfall methodology (Trivedi and Sharma 2013)*

This change in methodology was deemed to be necessary, as it felt more appropriate for the project as a whole when compared to agile and waterfall separately. Agile methodologies, while fitting for any software project, puts a focus on constant customer feedback during the SDLC (Hossain 2023), something that there would be a lack of in this project's case. The Waterfall methodology, while a lot simpler to plan with, its lack of flexibility when it comes to changes in project scope and lack of prototype creation in particular (ibid.), made it usage unfitting for the project. Combining parts of these methodologies however, made for a methodology that was fitting for the project as a whole. Though this was a unforeseen risk according to the risk matrix and risk register composed before the beginning of development (see Appendix B), this change did not negatively affect development, despite this shift occurring slightly into the execution of the project, as the project was able to stick to its defined timeline throughout the duration of the project execution (see Appendix C).

The Hybrid model was used by using a waterfall approach for the Requirements Analysis, Design, and Evaluation stage, and combining it with an iterative approach for the Implementation and Testing stage. The requirements were defined during the beginning of the project's execution stage within a Software Design Specification (see Appendix D), then, based on the requirements specified, designs surrounding those requirements were created.



Branch	Updated	Check status	Behind	Ahead	Pull request
clean-up	3 weeks ago		21	0	#2
req-1.2	3 weeks ago		30	0	
req-1.3	3 weeks ago		31	0	
req-1.6	last month		35	0	
req-1.4	last month		37	0	
req-1.1	last month		39	0	
req-1.5/1.7	last month		41	0	
notification-implement	last month		50	0	

Figure 2: *GitHub branches for each requirement*

Figure 1 showcases the process behind implementing and testing the PRI. When building prototypes of the PRI, a GitHub branch that matched the name of the requirement was created, then, within that branch, that requirement was implemented, then tested upon via Incremental Integration Testing to ensure the requirement was implemented as intended. If that part of the requirement passed all test cases as intended, a new GitHub branch for the next requirement was created using the newly completed requirement as a blueprint, and this process repeated until the implementation of all requirements were completed.

3.4 Requirements Analysis

3.5 Design

3.6 Implementation

3.7 Testing

At the end of each prototype iteration, Incremental Integration Testing was carried out to ensure that the requirements were implemented as intended according to how the requirement was defined in the SRS. Incremental Integration Testing can be defined as testing components of a software system, one by one, adding components together until eventually the entire system is being tested. As the nature of development involved creating a component based on the requirements specified, and only continuing once the component had been created to match the specification, this method of testing felt most appropriate.

Test ID	**Action**	**Expected Outcome**	**Actual Outcome**	**Evidence**	Assumptions / Context (If necessary)
2.1	User clicks on "previous week" button while no data for the previous week exists	A dialog pops up explaining to the user that no data for the previous week currently exists, and that they should check back later	A dialog pops up explaining to the user that no data for the previous week currently exists, and that they should check back later	Req_2.1.webm	

Table 2: Example of a test case

An example of how testing was carried out can be seen in Table 2. The test case includes the test ID, mapped to match the requirement number, the action taken by a user that would trigger the component, the expected outcome of the test, the actual outcome of said test, evidence to show the outcome of the test, usually in the form of a screen recorded video or screen capture, and any additional context or assumptions needed to understand the test case. A comprehensive list of the test cases and their results can be found in Appendix E.

4 Evaluation & Results Discussion

This section provides details on the evaluation of the project in relation to the hypothesis stated in section 1.3, which was:

The combination of usage tracking and intervention methods rooted in psychology within the created mobile application will help to address and reduce smartphone & social media usage in young adults.

It also gives critical analysis and discussion of the results and findings from the evaluation, before drawing initial conclusions based on the project hypothesis, research statement and objectives. Given the aforementioned hypothesis, the intention of the evaluation was to find if:

1. The mobile phone usage from a participant decreased when comparing their usage from the start of the evaluation to the end.
2. If there was a decrease in usage, was the PRI actually a contributing factor to it?

When evaluating if the PRI met the project's hypothesis, objective, and research statement, a mixed-method approach was used. (define mixed-method approach) The evaluation results are based off of the captured usage data from each participant, as well as the participant's own opinions of their experience using the PRI via a survey given at the end of the evaluation period. While the obtained usage stats will be the main focus, the results from the survey help to provide additional context and insights that quantitative data cannot easily do.

4.1 PRI-Evaluation results

Participants who agreed to take part in the evaluation were given the PRI for a period of 3 weeks. Alongside this, they were given a Quick Start Guide that gave extensive details regarding using the app, as well as instructions on how to install the app and troubleshooting advice. Once they had set up the application, they were also asked to send their usage immediately afterwards. This was to use said usage stats as a baseline to compare the usage for the next three weeks. While the data extracted initially shows the total usage which includes all apps used by a participant, the data was then further parsed by the author to analyse if there was a shift in usage of social media apps.

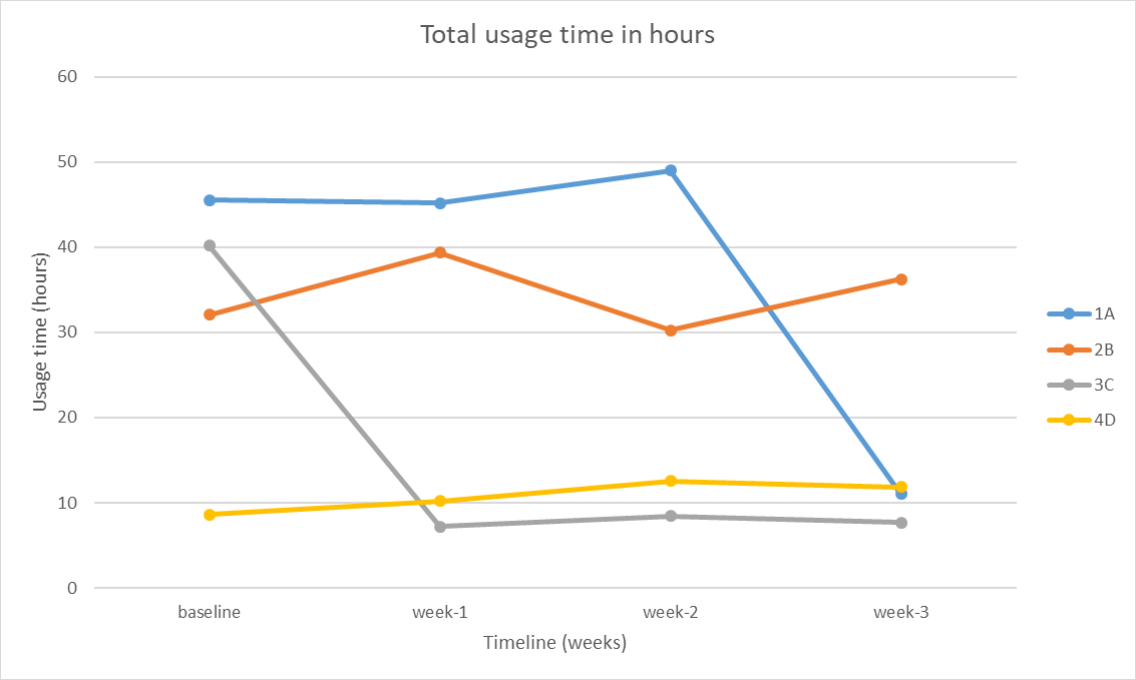


Figure 3: *Results from PRI evaluation*

Figure 2 shows the initial results from the evaluation, with the total usage from each participant. The x-axis shows the timeline of the evaluation period, from baseline to week 3. The y-axis shows the usage in hours. Table 3 shows a sample of the usage data captured, showing the name of the app package (unique identifier for android apps), the usage time for the app, and the percentage of the usage time.

android	24:50:00	61%
maps	04:18	10%
One UI Home	01:21	3%

Table 3: Sample of usage data from participant 3C.

Within the dataset, an outlier was detected within the baseline data for participant 3C. Seen above in table 3, the the "android" package, (referring to the home screen? quote a source) takes up over 24 hours of usage time. Seeing as this may be unrealistic for a typical user, the decision was made to leave out the usage time for "android". Figure 3 below shows the same chart as Figure 2, but without the outlier data.

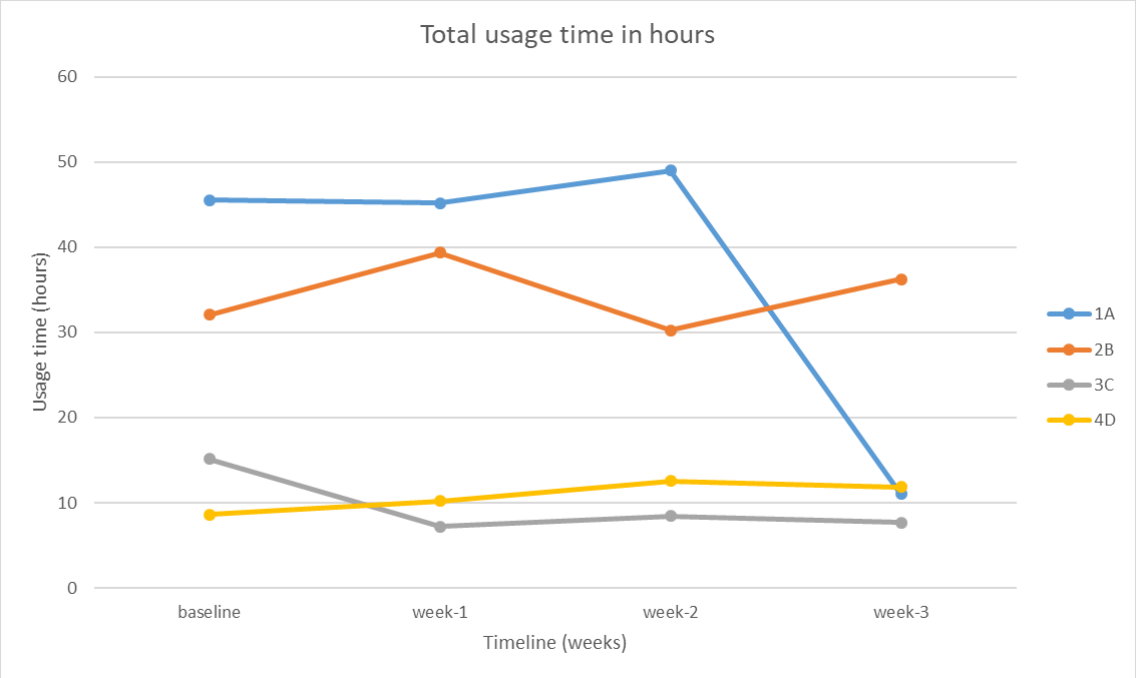


Figure 4: *Results from PRI evaluation, without the outlier data*

1A's usage is relatively consistent between baseline and week-2, though doesn't take a noticeable drop, but takes a substantial drop in week-3. This may be because... (expand on this, possibly due to how androids UsageStatsManager queries data) 2B's usage fluctuates a lot throughout the timeframe, initially rising before falling again, and then settling in the middle for week-3. 3C's usage does take a drop overall when comparing baseline to week-3, and remains under 10 hours for the remainder of the evaluation period after the baseline data. 4D's usage experiences a slight increase before dropping slightly, though experiences an increase when compared from the baseline to week-3.

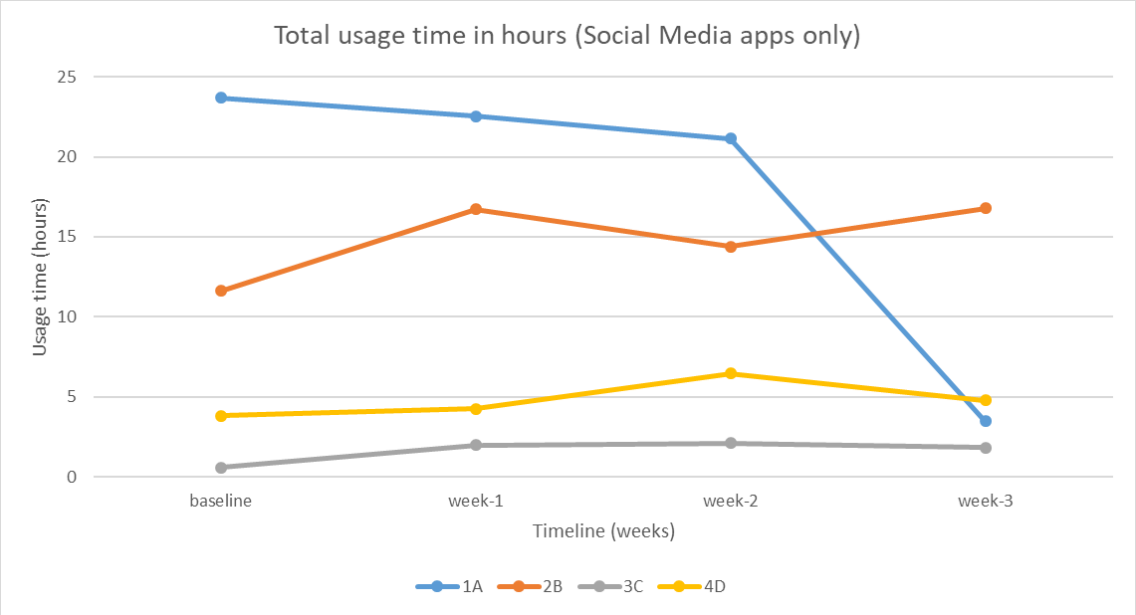


Figure 5: *Results from PRI evaluation, only showing social media usage*

Figure 4 shows the same graph as Figure 3, but only including social media usage from each participant. While the graph is relatively the same, there are some differences. For example, unlike the total usage, 1A's social media usage never increases, rather, taking a small decline instead. 4D's social media usage starts, and remains quite low, though does take a small increase in week-1 before levelling out by the latter weeks.

4.2 Post-Evaluation survey results

Part of using a mixed-method approach involves collected qualitative data, which can be defined as written text, images, audio recordings to name a few (Bernard, Wutich, and Ryan 2016). While the data from the previous section are an important piece of the evaluation as a whole, getting this kind of data helps to understand the thoughts and feelings from participants, and help in understanding and contextualising the behaviours represented in the previous section (Sutton and Austin 2015).

At the end of the evaluation testing period, participants were sent a survey to fill out. This survey gave participants the opportunity to express their opinions on using the app throughout the evaluation period, with the survey including a variety of multiple choice and open-ended, optional questions. Two question sets were provided, one focusing on how the participant felt after using the app, and the other focusing on the overall functionality of the app. The number of respondents

matched the number of participants who took part in the evaluation, that being 4. Note that because the open-ended questions were optional, not all participants provided a response to them.

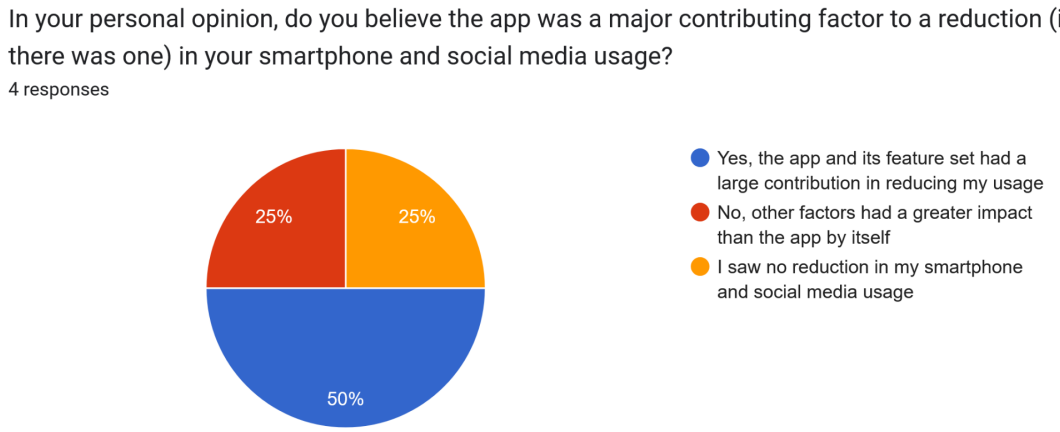


Figure 6: *Results from question 3 from the post-evaluation survey*

A complete collection of the responses and their associated questions can be found in Appendix F, however, the sentiment derived from the results of QS1 show that users felt more informed about their usage after using the app, and on average, users felt as if the app implemented the WDEP system to a good degree. As demonstrated in Figure 3, only two of the respondents felt that the app contributed to their lowered usage, with one respondent feeling as if other factors had a bigger impact than the app by itself, and the other feeling like there wasn't any reduction in their usage. When asked to provide a reason to their choice, one respondent said in reference to the app lowering their usage:

"It works pretty well! I feel as if seeing your values and hobbies increases motivation to not chase social media and other applications"

Another response, in reference to the participant not seeing a usage drop, stated:

"At this time of year as a student based outside the city of my university I was regularly on my device to check on my classes and assignments"

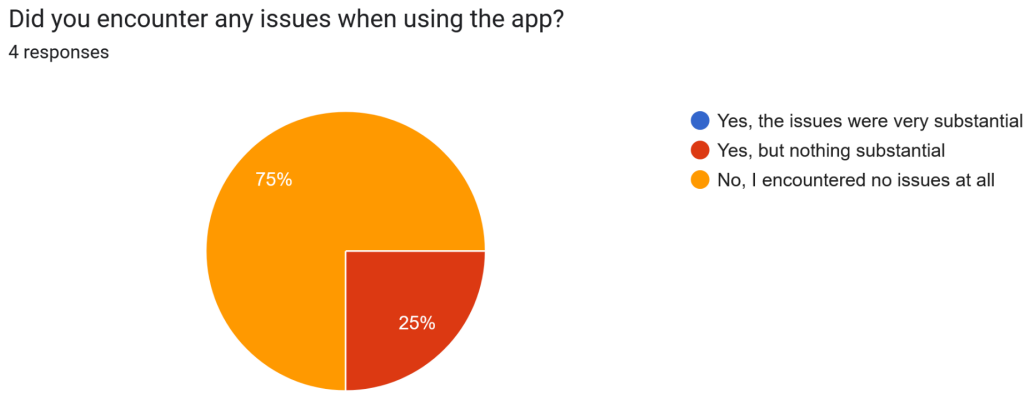


Figure 7: *Results from question 11 from the post-evaluation survey*

The responses to QS2 show that respondents generally had a positive experience using the app, with no participant experiencing any substantial issues. One participant did however note one smaller issue, stating:

"I find that the email prompt would be better and easier to understand is [sic] it was titled "report" or something similar"

Participants were asked if the app experience was better than existing apps that provided similar functionality, while 3 of the participants stated that they haven't used apps similar to the PRI, one participant disagreed that the app experience was better than existing apps. When asked to provide a follow-up stating what the PRI could take note of, they stated:

"an app called Screen Zen which gives more options to limit screen usage."

When asked to provide suggestions on how the app could be improved, one response suggested increasing the notification frequency from being sent on a weekly basis to a daily basis, while another suggested showing how the app usage for specific apps have changed between weeks. At the end, participants were finally asked to provide any closing thoughts regarding their experience, and while only one participant gave a response, they stated that they believe the app is a step in the right direction in addressing SA and SMA.

4.3 Discussion

5 Legal, Social, Ethical, and Professional Issues

This section will provide commentary on Legal, Social, Ethical, and Professional (LSEP) issues the project dealt with over its lifespan. Within the Social Issues section, it will also provide additional commentary and reflection on the Sustainable Development Goals (SDGs) the project has aligned itself with in accordance to the definitions provided by the United Nations.

5.1 Legal Issues

5.2 Social Issues

5.3 Ethical Issues

Arguably, the largest issue the project had to consider during its lifespan were Ethical Issues. As previously mentioned when discussing the project's primary objectives in section 1.2.2, for conduction the evaluation, human participants were required to carry out said evaluation as intended. This aspect by itself already constitutes as an ethical issue, given the use of human participants for a project of this nature means that..(citation here with consequence)...but this use of human participants brings forth another issue. The evaluation involved collecting app usage data from participants, as well as responses from the post-evaluation survey.

Similarly, during the requirements analysis stage of the project's execution, a survey was distributed, and 1-to-1 interviews were conducted. Given the use of software tools to carry out these parts of the project, this raises issues with data protection, informed consent, and the privacy of participants (Dave Pregoner et al. 2025). Dealing with such issues required taking different precautions at each step. When ensuring the privacy of participants at any stage, identifying information was either redacted, anonymised, or not collected at all. Alongside this, any data taken from participants was ensured to be compliant under the Data Protection Act 1998. When ensuring the protection of user data, all information was kept in a local folder on the author's PC at his home, as well as on an external USB. When conducting user interviews and the evaluation, participants were given a consent form to sign before taking part in anything related to the project, and before taking part in any survey, participants were asked to provide informed consent to take part and were given as much information as possible, something which was noted as an issue by Dave Pregoner et al. (ibid.).

To conduct any part of the project where human participants are required, ethical approval via an ethics form is required to be submitted prior to conducting any part

of it. All the details written about previously were mentioned within the ethics form, alongside reasonable justification, and ethical approval was granted.

5.4 Professional Issues

6 Conclusions, Project Critique, and Further Work

6.1 Project Critique

6.2 Further Work

6.3 Conclusion

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253BHuang%5C%252C+Shiqi%5C%253BWang%5C%252C+Xinman%5C%253BHan%5C%252C+Shasha%5C%253BZhu%5C%252C+Sui%5C%253BYoujing+Ning%5C%253BFang-fang%5C%252C+Zeng%5C%253BYuan%5C%252C+Yuan%5C&isbn=%5C&jtitle=Journal+of+Medical+Internet+Research%5C&btitle=%5C&rft_id=info:eric/%5C&rft_id=info:doi/10.2196%5C%252F59656%20https://doi.org/10.2196/59656.

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Appendix

Appendix A - GitHub Repository

The GitHub repository containing the source code for the PRI, as well as all other documentation related to the Honours Project, is available via:

https://github.com/LiamDee/honours-project-25-26

Appendix B - Risk Matrix & Risk Register

Risk Register

Risk ID	Risk Name	Risk Description	Category	Risk Likelihood	Risk Impact	Risk Response
R1	Unclear requirements	Requirements not made clear enough, leading to misunderstandings about said requirements	Clarity	Likely	4	Spend significant time researching similar products and projects to better understand what's usually needed.
R2	Poor set of information obtained from user interviews and surveys	Requirements and information obtained from users unclear, leading to a lacking set of features and user wants	Clarity	Unlikely	4	Ask a deep set of open-ended questions to invite detailed responses from the interviewee, promote surveys to a large set of relevant people.
R3	Lack of stakeholder engagement	Stakeholder(s) not engaged with the progression of the project	Clarity	Unlikely	3	Meet with stakeholders on a frequent basis to keep them informed and invested in the project, keep most meetings brief but informative, and save in-depth meetings for when milestones are reached.
R4	Stakeholder design disagreements	Stakeholder(s) have disagreements with one or more aspects about the design specifications	Organisational	Probable	2	Meet with all stakeholders whenever possible to understand points of contention, gather feedback, and come to an agreement on solutions in order to move past them.
R5	Technical issues	Technical problems arise during the implementation stage of development, leading to delays	Technical	Likely	4	Take time to understand the tools utilised to carry out development.
R6	Bad set of test cases	Test cases fail to properly test important aspects of the prototype	Technical	Unlikely	3	Look into how successful test plans are conducted, make sure test cases cover all the important aspects of the prototype at the very least.
R7	Lack of meaningful user feedback	Little feedback regarding what worked well or any improvements to the finished product that can be made	Clarity	Probable	2	Follow the response set out in risk R2, keep surveys concise to incentivise users to respond without taking up too much of their time.
R8	Missed deadlines	Deadlines defined aren't met, which could lead to aspects of the project being rushed to account for time	Time	Probable	5	Follow the deadlines set out as closely as possible, getting assistance if necessary.

Figure 8: *Risk Register (originally from RS&PI Project Planning report)*

Risk Matrix

Risk Matrix

LIKELIHOOD \ SEVERITY	1 NEGLIGIBLE	2 MINOR	3 MODERATE	4 MAJOR	5 CATASTROPHIC
5 - VERY LIKELY	RESCHEDULE MEETING		INEXPERIENCE WITH SOFTWARE & ASSOCIATED TOOLS		
4 - LIKELY		SCOPE CREEP	ADDITIONAL RESEARCH NEEDED AFTER ANALYSIS STAGE	LACK OF MOTIVATION / BURNOUT	UNDERESTIMATION OF REQUIRED RESOURCES
3 - PROBABLE	ILLNESS		DELAY IN REACHING MILESTONES	TIME CRUNCH	
2 - UNLIKELY		LACK OF USER / STAKEHOLDER ENGAGEMENT		MAJOR INJURY	FILES ARE CORRUPTED DURING DEVELOPMENT
1 - VERY UNLIKELY	SHORT POWER OUTAGE	OPERATIONAL MISHAPS			TECHNOLOGY BREAKS DOWN

Figure 9: *Risk Matrix (originally from RS&PI Project Planning report)*

Appendix C - Gantt Chart

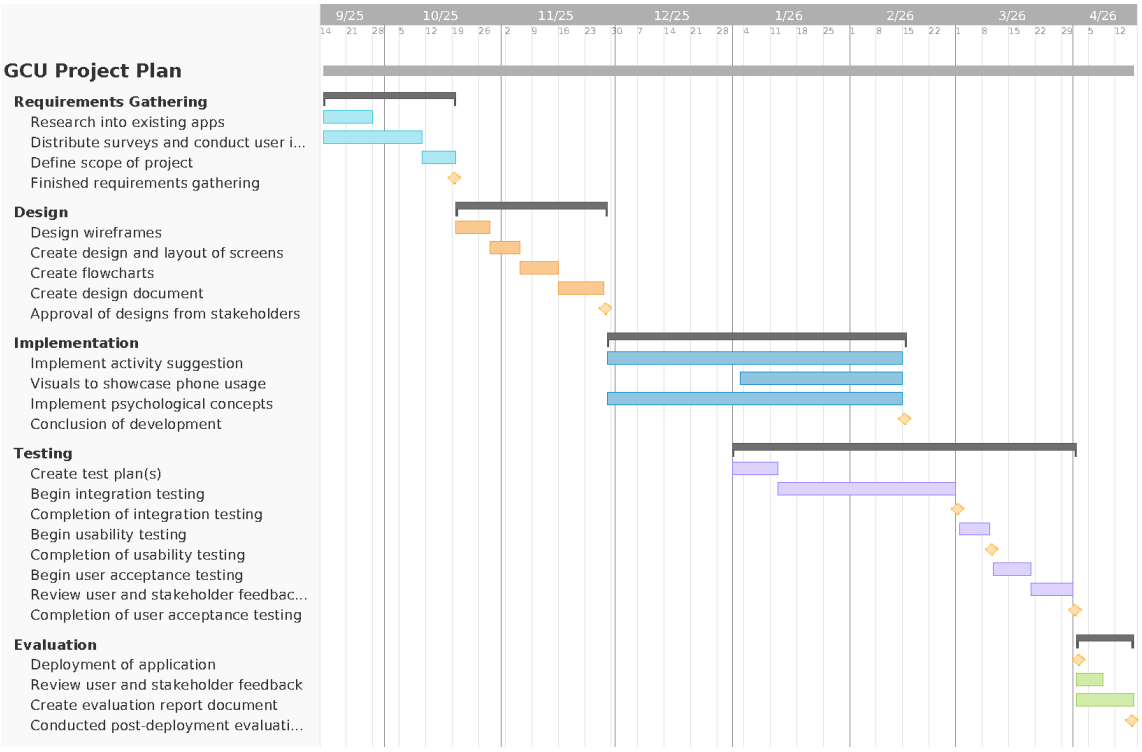

Figure 10: *Project Plan Gantt Chart (originally from RS&PI Project Planning report)*

Appendix D - Software Requirements Specification

Appendix E - Test Case results

Appendix F - Post-Evaluation survey results